

Hiren Madhu

Research interests: Foundation Models, Representation Learning, Geometric Deep Learning

 <https://hirenmadhu.github.io>

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 [Google Scholar](#)

 [HirenMadhu](#)

Education

Yale University

PhD in Computer Science, GPA 4.0 / 4.0

2024-2029

Co-advisors: Smita Krishnaswamy, Rex Ying

Indian Institute of Science, Bangalore

MTech in Artificial Intelligence, GPA 8.1 / 10

2021-2023

Advisor: Sundeep Chopuri

LDRP-ITR, Gandhinagar

BE in Computer Engineering, GPA 8.4 / 10

2017-2021

Co-advisors: Sandip Modha, Thomas Mandl

Research Experience

NetApp

June 2026 – Aug. 2026

AI Intern (Mentored by Kiran Srinivasan)

San Jose, CA

- Building a system where an LLM, fine-tuned with LoRA, answers natural-language questions about historical relational data grounded entirely in a specialized encoder's embeddings, with no forecasting done by the LLM.
- Projected the encoder's row-token embeddings into the LLM token space as complete context with no retrieval step, a retrieval-free alternative to RAG for grounding the model in structured data.
- Built the upstream model as a relational mixture-of-experts encoder (MoRE) that forecasts over cross-schema time-series tables, with a learned router over numeric, text, categorical, and joint-schema experts.

Indian Institute of Science, Bangalore

June 2023 – June 2024

Pre-doc Research Fellow (Advised by Prof. Sundeep Chopuri)

Bangalore, IN

- Developed weakly-supervised representation learning techniques for topological structures.
- Designed an augmentation method and contrastive loss for simplicial complexes, achieving higher performance and robustness compared to supervised baselines (NeurIPS 2023).
- Proposed a parameter-free scattering framework for unsupervised feature extraction on simplicial complexes, improving benchmark performance and efficiency (ICML 2024).

Publications

Preprints and Conference Proceedings (* equal contribution)

- ImmunoFoundation: A Multimodal Foundation Model for Immunogenicity Prediction and Peptide Optimization.** João Felipe Rocha*, **H. Madhu***, et al. **FM4LS Workshop @ ICML 2026**. [OpenReview]
- HypRAG: Hyperbolic Dense Retrieval for Retrieval Augmented Generation.** **H. Madhu**, et al. **ICML 2026**. [Arxiv]
- HEIST: A Graph Foundation Model for Spatial Transcriptomics and Proteomics Data.** **H. Madhu**, et al. **ICLR 2026**. [Arxiv]
- HELM: Hyperbolic Large Language Models via Mixture-of-Curvature Experts.** N. He*, R. Anand*, **H. Madhu**, et al. **NeurIPS 2025**. [NeurIPS] [Arxiv]
- HiPoNet: A Topology-Preserving Multi-View Neural Network For High Dimensional Point Cloud and Single-Cell Data.** S. Viswanath*, **H. Madhu***, et al. **NeurIPS 2025**. [NeurIPS] [Arxiv]
- Hyperbolic Deep Learning for Foundation Models: A Survey.** N. He, **H. Madhu**, et al. **KDD 2025 Tutorial**. [ACM]
- Unsupervised Parameter-free Simplicial Representation Learning with Scattering Transforms.** **H. Madhu***, S. Gurugubelli*, SP Chopuri. **ICML 2024**. [ICML]
- TopoSRL: Topology Preserving Self-Supervised Simplicial Representation Learning.** **H. Madhu**, SP Chopuri. **NeurIPS 2023**. [NeurIPS]
- Detecting offensive speech in conversational code-mixed dialogue on social media: A contextual dataset and benchmark experiments.** **H. Madhu**, et al. **Expert Systems with Applications**. [Elsevier]

Projects

HypReason | LLMs, Post-Training, Reinforcement Learning | Graph-and-Geometric-Learning

Apr 2026 – Now

- Developing a **hyperbolic post-training** pipeline for LLM reasoning, projecting pretrained representations onto the **Lorentz hyperboloid** to exploit the latent tree structure of chain-of-thought reasoning.
- Built a **hyperbolic reward model** and adapted **GRPO** to Riemannian optimization, with HypLoRA-based SFT and learnable per-layer curvature for hierarchical reward shaping.
- Benchmarking on GSM8K, MATH-500, AIME, and GPQA Diamond against Euclidean baselines under matched compute, targeting **accuracy and inference-efficiency** gains at smaller parameter counts.

Technical Skills & Recognition

Frameworks: PyTorch, PyTorch Geometric, Huggingface, DeepSpeed/FSDP

Languages: Python · **Systems:** distributed / multi-GPU training, mixed precision

Recognition: Top Reviewer NeurIPS 2025 · GATE CS 2021 All India Rank 87 (Score 860)